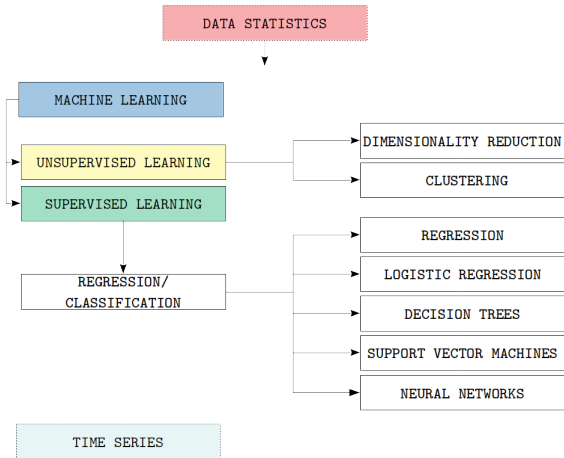


- linear/polynomial regression
 - ▶ 'An Introduction to Statistical Learning', Sections 3.1-3.3, 7.1
- logistic regression
 - ▶ 'An Introduction to Statistical Learning', Section 4.3
- decision trees
 - ▶ 'An Introduction to Statistical Learning', Section 8.1
- support vector machines
 - ▶ 'An Introduction to Statistical Learning', Sections 9.1-9.3
- neural networks
 - ▶ 'An Introduction to Statistical Learning', Sections 10.1-10.2
 - ▶ 'The Elements of Statistical Learning', Section 11

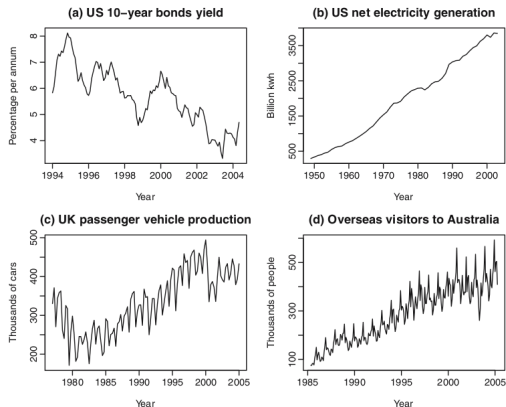
Overview

- 1 Introduction
- 2 Basic statistics
- 3 Multivariate statistics and Principal Component Analysis
- 4 Unsupervised learning: Clustering
- 5 Supervised learning
- 6 Overview on Time Series



what is a time series

”**ordered** sequence of values of a variable observed at equally spaced time instants”



two main methodologies:

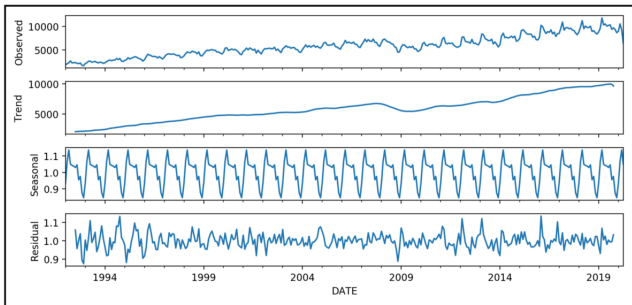
- **time series analysis:** analyzing time series data to extract knowledge
- **time series forecast:** build models to predict future values based on past

for each time instant t , time series y_t is given by a **signal** x_t and a **residual** noise ϵ_t as

$$y_t = x_t + \epsilon_t \quad \xrightarrow{\text{WHITE NOISE}} \approx \mathcal{N}(x)$$

signal x_t can be decomposed into multiple underlying signals including:

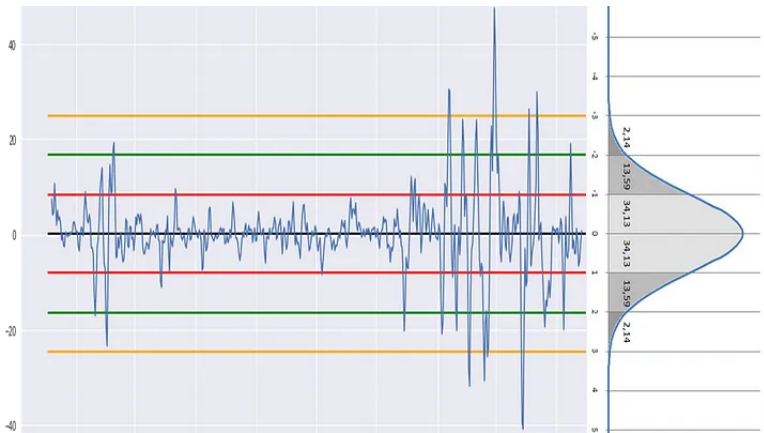
- **Trend (T)**: the long-term direction of the series
- **Seasonal (S)**: a pattern that repeats with a known periodicity



- **additive** model: $y = (T + S) + \epsilon$
- **multiplicative** model: $y = (T \times S) + \epsilon$

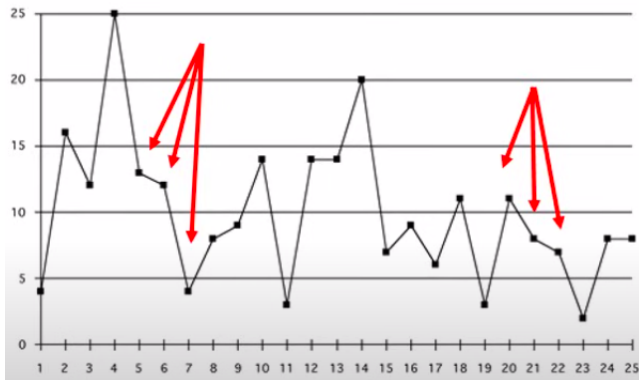
t.s. share some aspects with random variables and probability distributions...

- may be described by **statistical moments** (e.g. mean and standard deviation)



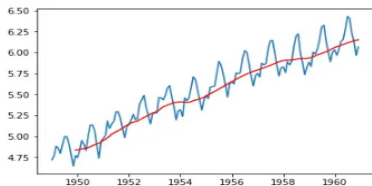
N.B. t.s. is not a sequence of extractions from a probability distribution!

- t.s. has **memory**: y_t is influenced by previous values $y_{t-1}, y_{t-2}, \dots$
↳ LAGGED VALUES

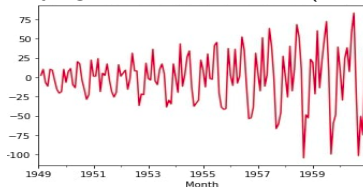


- statistical moments may vary through time

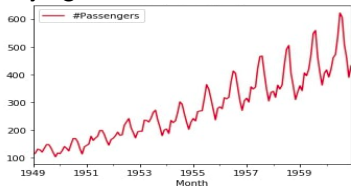
varying mean (trend)



varying standard deviation (**volatility**)

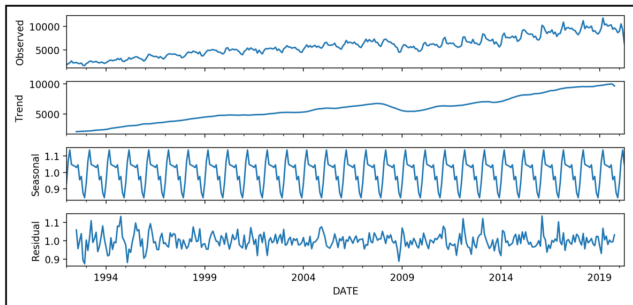


varying both



t.s. share some aspects with (linear) regression...

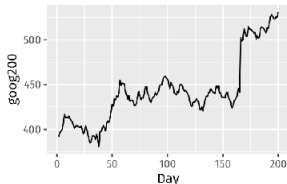
- like regression, t.s. analysis focuses on describing mathematically the process underlying data and ultimately on making predictions
- linear regression quantifies the specific impact of the predictors on the output as $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$, while t.s analysis decomposes the underlying process into signals (trend, seasonal,...) and noise to combine them to replicate the process



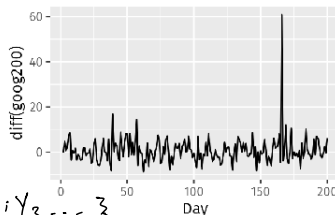
stationarity vs non-stationarity

in **stationary** t.s. the mean (**trend stationary**) and standard deviation are constant

non-stationary
(google stocks price)



stationary
(daily variation of google stocks price)

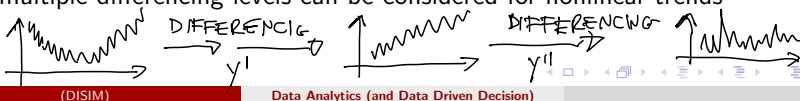


differencing t.s.: $y'_t = y_t - y_{t-1}$

$$y_t = \{y_1, y_2, y_3, \dots\}$$

$$y'_t = \{y_2 - y_1, y_3 - y_2, y_4 - y_3, \dots\}$$

- non-stationary t.s. with sufficiently linear trend can be made stationary by differencing
- multiple differencing levels can be considered for nonlinear trends



autocorrelation

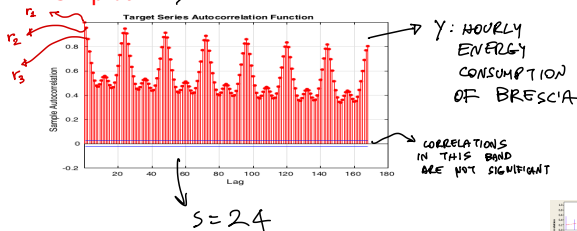
t.s. has memory $\implies$ current value y_t influences subsequent **lagged** values y_{t+l}

autocorrelation function (ACF) measure the overall correlation between values observed at distance (lag) ℓ , i.e. y_t and y_{t+l}

• **ACF** on N t.s. observations: $r_\ell = \frac{\sum_{i=1}^{N-\ell} (y_i - \bar{y})(y_{i+\ell} - \bar{y})}{\sum_{i=1}^N (y_i - \bar{y})^2} \rightarrow r_\ell \in [-1, 1]$

AVERAGE OF $\bar{y}$ OVER N

• **ACF plot:** *PLOT r_ℓ FOR MULTIPLE ℓ*



- non-stationary t.s. ACF plot goes to zero fast
- a seasonality value s can be detected observing ACF plot
- **seasonal differencing**, i.e. $y'_t = y_t - y_{t-s}$, to check if t.s. is a **seasonal naive**

TRIVIAL
REPETITION
OF THE
PROCESS
OVER
PERIODS



Time Series forecasting

autocorrelation can be used to drive the construction of forecasting **linear models** in which future t.s. values are linear combination of past observations

- **AR(p)** (autoregressive) model: the t.s. forecast is obtained as a linear combination of p (AR order) past t.s. observations as

$$y_t = \sum_{i=1}^p \phi_i y_{t-i} + \epsilon_t$$

AR COEFFICIENTS

ERROR AT TIME t

- **MA(q)** (moving average) model: the t.s. forecast is obtained as a linear combination of q (MA order) past forecasting errors as

STATIONARY TIME-SERIES

$$y_t = \mu + \sum_{i=1}^q \theta_i \epsilon_{t-i} + \epsilon_t$$

MA COEFFICIENTS

AVERAGE

- **I(d)** (integrated): differencing is applied d times to the t.s. to be modeled

$$\text{ARIMA}(p,d,q): y_t^{I(d)} = \mu + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{i=1}^q \theta_i \epsilon_{t-i} + \epsilon_t$$

Time Series metrics

how to evaluate the accuracy of a t.s. forecasting model?

standard regression metrics like MSE , $RMSE$, R^2 can be used but they do not represent the real impact of the error...

SCALE-DEPENDENT

- $MAPE$ (mean absolute percentage error): $100 \frac{1}{N} \sum_{t=1}^N \frac{|y_t - \hat{y}_t|}{|y_t|}$
 - ▶ $MAPE$ may overestimate errors for small y_t ($MAPE = \infty$ for $y_t = 0$)

REAL VALUES
PREDICTED VALUES

- $sMAPE$ (symmetric $MAPE$): $100 \frac{1}{N} \sum_{t=1}^N \frac{|y_t - \hat{y}_t|}{\frac{|y_t| + |\hat{y}_t|}{2}}$
 - ▶ $sMAPE$ fixes the drawback of $MAPE$ as it is comprised in 0% and 200%
 - ▶ it can be biased towards high errors for small values of one of y_t or $\hat{y}_t$

WHEN BOTH y_t AND $\hat{y}_t = 0$
 $sMAPE$ IS NOT DEFINED

$$\begin{aligned} y_t &= 100 \\ \hat{y}_t &= 110 \end{aligned} \Rightarrow \begin{aligned} MAPE &= \frac{10}{100} = 0,1 \\ sMAPE &= \frac{10}{105} \end{aligned}$$

$MAPE > sMAPE$

$$\begin{aligned} y_t &= 100 \\ \hat{y}_t &= 90 \end{aligned} \Rightarrow \begin{aligned} MAPE &= \frac{10}{100} = 0,1 \\ sMAPE &= \frac{10}{95} \end{aligned}$$

$MAPE < sMAPE$

exercise 5.1

